Harshith M

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Professional Summary

Full-Stack Developer and 3rd-year Computer Science student at BMS College of Engineering, experienced in building and deploying real-world web applications using Next.js, React, TypeScript, and MongoDB. Strong foundation in Data Structures & Algorithms (C++) with a focus on scalable backend systems.

Experience

Pranasya Technologies — Full Stack Developer *Bangalore · Aug 2026 – Present*

Bharat Quest AI — offline-first classroom learning platform *Python · React · Three.js · Fastify · PostgreSQL*

- Built a six-stage Python RAG pipeline turning NCERT Class 5–8 textbooks into a validated question bank: pypdf extraction with page tracking, 500-character chunking at 150-character overlap, all-MiniLM-L6-v2 embeddings indexed per class in ChromaDB.
- Gated every generated question against its source chunk at an 80% grounding threshold, deduplicated semantically at 0.85 cosine similarity, and mapped questions to a 30-state gazetteer before a seeded, answer-position-balanced export — so nothing reaches a classroom ungrounded in the textbook.
- Shipped the client as an offline-first React app for classroom smartboards: IndexedDB persistence with backend reconciliation on reconnect, JWT auth with an admin panel, and an interactive 3D India map in react-three-fiber and GSAP.

WhatsApp Alert & Broadcast Platform *React · TypeScript · Node/Express · Prisma · MySQL · Docker*

- Built a role-aware internal broadcast platform end to end: four-tier RBAC from super-admin to viewer with per-role PII phone masking, and a draft → approval → queue campaign workflow.
- Engineered the delivery engine as a background worker using transactional claim locks (claim token + worker ID + TTL) so concurrently scaled workers never double-send, with 30s/2m/10m exponential backoff, automatic stale-claim recovery after crashes, and graceful SIGINT/SIGTERM drain.
- Hardened the API with HttpOnly-cookie JWT auth, CSRF tokens, rate limiting, Zod validation, SHA-256 request-hashed idempotency keys, and audit logging; gated merges on GitHub Actions CI running typecheck plus unit and end-to-end suites against a MySQL service container.

Projects

Quantro — AI Stock Portfolio Manager *Next.js · PostgreSQL · AWS Lambda · S3 · JWT · Claude API*

- Developed an AI-driven portfolio management platform for Indian investors with portfolio tracking, trade simulation, backtesting, watchlists, and personalized stock recommendations across 50+ NSE stocks.
- Engineered a backend market-data pipeline using GitHub Actions, AWS Lambda, and Amazon S3 to fetch, store, and process end-of-day OHLCV data for automated portfolio and analytics updates.
- Implemented secure authentication and user management with JWT, and integrated the Claude API to generate personalized stock recommendations based on user risk profile and portfolio context.

Sniply — URL Shortener with Analytics *Next.js · MongoDB · Clerk*

- Non-blocking 307 redirect path with asynchronous write-behind logging of browser, OS, device, and referrer; cut redirect latency ~20% via query optimisation and indexing.

Certifications

AWS Certified Cloud Practitioner (AWS CCP) *2026*

Achievements & Positions of Responsibility

1st Place — CODEVERSE 24-hour Hackathon, BMSCE IEEE Computer Society, college-wide. *2025*

Student Activity Coordinator — BMSCE IEEE Student Branch *2025*

Education

B.M.S. College of Engineering, Bangalore *2024 – 2028*

B.E. Computer Science & Engineering · CGPA **8.92 / 10**

Technical Skills

Languages	TypeScript, JavaScript, Python, C++, SQL
Backend	Node.js, Express, REST API design, Prisma, JWT, RBAC
Frontend	React, Next.js, Vite, Tailwind CSS, Three.js / react-three-fiber
Data & AI	PostgreSQL, MySQL, MongoDB, ChromaDB, Claude API, Gemini API
Infra & Tooling	AWS (EC2, S3, Lambda, IAM), Docker, Git, Vercel